EE2004 Digital Signal Processing

Tutorial 9 Answer Key

1. $h[n] = h^*[N-1-n]$
2. (a) not generalized linear phase, (b) generalized linear phase
3. Remaining zeros: -2, $0.3 - j\,0.5$, $-\frac{1}{2} - j\frac{\sqrt{3}}{2}$, $0.88235 \pm j\,1.47059$.
   $H(z) = 1 + 1.13529\,z^{-1} + 0.56353\,z^{-2} + 5.68412\,z^{-3} + 4.97706\,z^{-4} + 5.68412\,z^{-5} + 0.56353\,z^{-6} + 1.13529\,z^{-7} + z^{-8}$
4. (a) $g[n] = \delta\left[n - \frac{M}{2}\right] - h[n]$, Type I
   (b) $H_{ZP}(\omega) = 1 - G_{ZP}(\omega)$
   (c) $F_{ZP}(\omega)$ is highpass that is approximately zero for $0 \leq \omega \leq \frac{2\pi}{3}$ and approximately one for $\frac{3\pi}{4} \leq \omega \leq \pi$.
5. (a) $2\sin(1.5\omega) + 2\sin(0.5\omega)$
   (b) $2 - 4\cos(\omega) + 2\cos(2\omega)$
6. $h[n] = \{-0.16,\ 0.32,\ 0,\ -0.32,\ 0.16\}$
7. $\{3,\ 1,\ 2,\ 0,\ -2,\ -1,\ -3\}$